Menna Ewis

AI Research Engineer | Computer Vision & LLM Systems | IEEE Published Researcher
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PROFESSIONAL EXPERIENCE

AI Automation Engineer *Nov 2025 – Present*
Qanawat.ai — Remote, Kuwait

- Designed and deployed enterprise-grade AI systems integrating LLMs, RAG pipelines, vector databases, and agentic multi-agent workflows for large-scale medical and operational datasets.
- Built intelligent conversational systems with hybrid retrieval architectures, optimizing retrieval performance and reducing query latency by 30%.
- Implemented multi-agent orchestration workflows with stateful execution and external tool integrations.
- Automated reporting pipelines using n8n, Google Apps Script, and Google Workspace APIs.

AI Systems Developer *Feb 2024 – Oct 2025*
IndegreeIT — Remote, Qatar

- Designed and deployed LLM-powered enterprise knowledge retrieval systems and AI-driven automation pipelines.
- Built RAG pipelines using dense vector embeddings and Elasticsearch for scalable semantic search.
- Developed scalable serverless microservices on AWS and integrated ML services into production backend systems.

AI Research & Development Engineer *Mar 2022 – Jul 2023*
Orange

- Conducted R&D on AI and ML solutions; developed and benchmarked machine learning models for data analysis and predictive systems.
- Participated in experimentation, benchmarking, and validation of AI solutions across multidisciplinary teams.
- Contributed to technical documentation, research deliverables, and internal knowledge sharing.

Backend Software Engineer *Aug 2023 – Feb 2024*
Halan-MNT

- Developed scalable backend services using Go and gRPC within a high-throughput distributed financial platform.
- Contributed to data pipeline design and microservice-based architectures improving service reliability.

Teaching Assistant – Computer Science *Aug 2023 – Mar 2024*
University of Prince Edward Island (UPEI), Canada

- Instructed Python programming, OOP, data structures, and algorithms to 200+ undergraduate students.
- Led lab sessions on backend programming, software engineering fundamentals, debugging, and best practices.

External Instructor (Part-Time) *Aug 2024 – Aug 2025*
Information Technology Institute (ITI)

- Delivered hands-on courses in Python and Django backend development to 100+ students.
- Covered REST API design, ORM optimization, authentication, and scalable software architecture.

KEY PROJECTS

Temporary Email Detection System | *DistilBERT • NLP • Explainable AI • Transfer Learning*

- Developed a hybrid classification system combining transformer-based NLP (DistilBERT) with domain-specific rule validation, achieving high-precision detection of disposable email domains.

- Applied LIME explainability to interpret model decisions — directly applicable to transparent AI systems in safety-critical contexts such as autonomous vehicles.

UChat — MCP AI Orchestration Server | *LLMs • Multi-Agent Systems • Event-Driven Architecture*

- Designed a Model Context Protocol (MCP) server for multi-agent orchestration with tool routing, session management, and complex workflow execution logic.
- Integrated Vision-Language Model (VLM) capabilities and external APIs to support multimodal AI-driven workflows.

HR.AI Platform | *Django • PostgreSQL • AWS • ML Integration*

- Built backend services for workforce analytics, designing data models and REST APIs integrating ML model outputs into production workflows.

TECHNICAL SKILLS

Deep Learning	PyTorch, TensorFlow, Keras, Scikit-Learn, Transfer Learning, Model Evaluation — certified: TensorFlow on GCP (Intermediate ML), Image Processing ML, Language Processing ML
Computer Vision	OpenCV, CNNs, Object Detection, Image/Video Processing, Scene Understanding
NLP & LLMs	Transformers (HuggingFace), RAG, Prompt Engineering, VLMs, AI Agents — certified: Introduction to Large Language Models, IBM Mastering the Art of Prompting
Languages	Python (expert), Go, JavaScript, SQL, PySpark — certified: Go Essentials
Big Data & MLOps	Apache Spark, Azure Databricks, Docker, Git, CI/CD — certified: Git Essential Training, Advanced Data Analysis (Udacity)
Cloud Platforms	AWS, Azure ML, Azure Databricks, Serverless Architectures — certified: AWS for Developers
Databases	PostgreSQL, Elasticsearch, Vector Databases, SQL Server
Backend	Django, FastAPI, Flask, REST APIs, gRPC, Microservices — certified: Manara Certified Software Engineer
Research Skills	Experiment Design, Benchmarking, Performance Analysis, Technical Writing

EDUCATION

Master of Science (M.Sc.) in Computer Engineering *2025-present*

Faculty of Engineering Cairo University

B.Sc. in Computer Engineering *2018 - 2023*

Egypt Japan University of Science and Technology, GPA 3.4